

SEASONAL & REGIONAL EVALUATION OF SOLAR RADIATION MODELS

A Comprehensive Study Guide & Technical Project Report

Project Briefing: Assessing the geographical transferability and seasonal robustness of sixteen empirical temperature-based solar radiation models against the sunshine-based Angström–Prescott reference standard using 22 years of meteorological records (2002–2023) across India.

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STUDY GUIDE OUTLINE

Chapter 1: The Solar Resource & Network Constraints	Page 2
Chapter 2: Project Methodology & Mathematical Blueprint	Page 3
Chapter 3: Regional Diversity & Performance Evaluation	Page 4
Chapter 4: Stratified Framework & Agricultural Deployment	Page 5

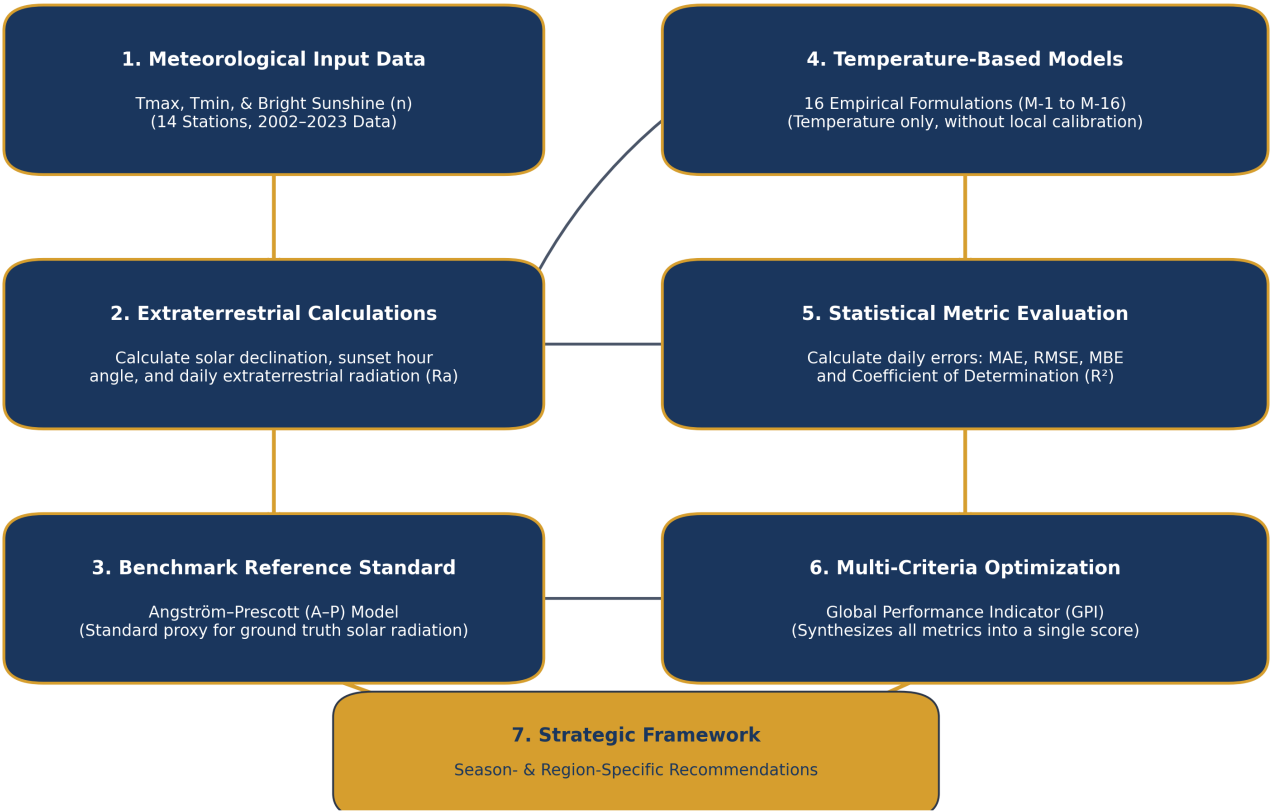
CHAPTER 1: THE SOLAR RESOURCE & NETWORK CONSTRAINTS

Incoming surface solar radiation is the driving engine of the earth's climate system and the primary determinant of crop development and agricultural productivity. It governs vital thermodynamic exchanges, soil evaporation, and crop transpiration (collectively known as **evapotranspiration**). Accurate estimation of incoming solar radiation is therefore essential for crop growth simulations, irrigation scheduling, hydrological modeling, and regional agricultural planning.

The Pyranometer Bottleneck: Despite its criticality, directly measuring solar radiation requires highly specialized radiometric sensors called **pyranometers**. These sensors are expensive to deploy, demand continuous maintenance, and require delicate calibrations. Consequently, ground-based pyranometer networks are sparse. The **India Meteorological Department (IMD)** operates only **46 solar radiation network stations** across all of India. In contrast, standard meteorological parameters—namely maximum (T_{max}) and minimum (T_{min}) temperatures—are monitored daily at over 770 surface weather and agrometeorological observatories across the country.

The Temperature-Based Solution: Under these network limitations, empirical formulas that can estimate solar radiation using *only* widely available daily temperature extremes act as the ultimate low-cost, high-reliability alternative. The physical foundation of these models lies in the **Diurnal Temperature Range** ($\Delta T = T_{max} - T_{min}$). Clear skies result in heavy daytime solar heating (high T_{max}) and rapid nighttime radiative cooling (low T_{min}), creating a wide ΔT . Cloudy days damp this variation, narrowing ΔT . This research provides the first nationally consistent, uncalibrated evaluation of 16 temperature-based models across 14 distinct climatic regions in India to develop an actionable, stratified selection guide.

FIGURE 1.1: IMD RESEARCH METHODOLOGY & EVALUATION PIPELINE



Visual Flowchart: Steps 1 to 7 representing the data collation, standard calibration, validation, and stratification.

CHAPTER 2: PROJECT METHODOLOGY & MATHEMATICAL BLUEPRINT

To establish a robust standard proxy for validation in the absence of absolute pyranometer readings, the researchers utilized the **Angström–Prescott (A–P) model**. The A–P model (Eq. 2.1) is mathematically robust, physically grounded, and uses sunshine duration as a direct proxy for cloud-induced solar attenuation:

Reference Standard: Angström–Prescott Model (Eq. 2.1)

$$R_s = [a + b * (n / N)] * R_a$$

Where: R_s is global solar radiation (MJ/m²/day); R_a is astronomical extraterrestrial radiation; n is the actual bright sunshine hours; N is the maximum possible astronomical sunshine hours; a and b are physical coefficients established at 0.25 and 0.50 respectively.

The study then evaluated 16 distinct temperature-based models. These empirical formulas are computationally clean, requiring *no sunshine data* or expensive equipment. This makes them ideal for automated agricultural weather stations deployed across remote rural regions. Six of the most prominent models are highlighted in Table 2.1:

Model	Developer & Reference	Mathematical Formulation
M-1	Hargreaves & Samani (1985)	$R_s = 0.16 * \sqrt{T_{max} - T_{min}} * R_a$
M-2	Rao et al. (2017)	$H / H_0 = 1.29 + 0.3626 * (T_{max} - T_{min})$
M-3	Richardson (2018) / Marif (2022)	$H / H_0 = 0.3124 * (\Delta T)^{0.2942}$
M-5	Chen et al. (2004)	$H / H_0 = 0.2063 + 0.1801 * \ln(\Delta T)$
M-11	Sivamadhavi & Selvaraj (2012)	$H / H_0 = 0.4610 + 0.0029 * \Delta T$
M-13	Hassan et al. (2016)	$H / H_0 = 0.206590 + 0.026851 * T_{max} - 0.024184 * T_{min}$

CREATIVE BLOCKS: THE TWO MASTER FORMULATIONS

CHAMPION MODEL 1: Hargreaves–Samani Formulation (M-1)

$$R_s = 0.16 * \sqrt{(T_{max} - T_{min})} * R_a$$

The Inland Giant: A beautifully simple model based entirely on the square-root of the diurnal temperature range. Highly effective during clear-sky, low-humidity seasons across the vast interiors of India.

CHAMPION MODEL 2: Rao et al. Monsoon Formulation (M-2)

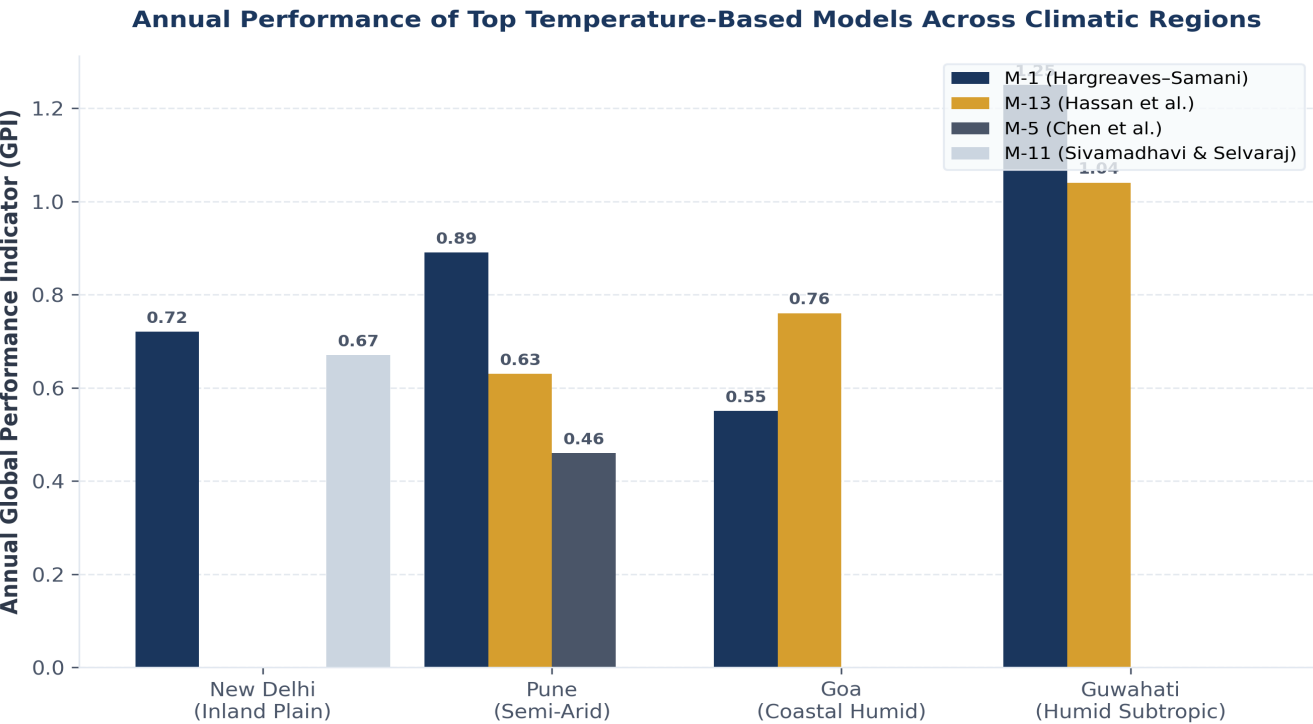
$$H / H_0 = 1.29 + 0.3626 * (T_{max} - T_{min})$$

The Cloud Specialist: Re-engineered to withstand cloud cover dynamics. By introducing structured linear scaling, it successfully counters the temperature-radiation decoupling during India's heavy monsoon months.

CHAPTER 3: STATISTICAL EVALUATION & REGIONAL DIVERSITY

To determine the best-performing models without bias, the study used 4 strict metrics: **MAE** (Mean Absolute Error), **RMSE** (Root Mean Square Error), **MBE** (Mean Bias Error), and the **Coefficient of Determination (R^2)**. Because different metrics often favor different models, the researchers calculated a synthesized **Global Performance Indicator (GPI)**. The GPI normalizes all 4 indicators and measures a model's deviation from the median performance. A positive, elevated GPI represents superior performance across all criteria.

FIGURE 3.1: ANNUAL GPI COMPOSITION BY REGIONAL CLIMATES



Performance Synthesis: This chart highlights how the optimal model changes between continental plains, semi-arid plateaus, humid coastal areas, and rainy eastern regions.

TABLE 3.1: REPRESENTATIVE AGROMETEOROLOGICAL OBSERVATORIES

ID	Observatory District	Latitude	Longitude	Alt (m)	Data Period
1	Bhubaneswar, Odisha	20.15° N	85.52° E	25.9	2002–2023 (22y)
2	Ela-Goa, Goa	15.30° N	73.55° E	60.0	2002–2023 (22y)
3	Guwahati, Assam	26.35° N	90.52° E	105.0	2002–2020 (19y)
4	Hyderabad, Telangana	17.19° N	78.23° E	520.0	2002–2023 (22y)
5	Jabalpur, Madhya Pradesh	23.13° N	79.57° E	411.0	2002–2023 (22y)
6	Jaipur, Rajasthan	26.51° N	75.47° E	390.0	2002–2023 (22y)
7	Jodhpur, Rajasthan	26.18° N	73.01° E	224.0	2002–2023 (22y)
8	New Delhi, Delhi	28.04° N	77.10° E	220.0	2003–2023 (21y)
9	Pune, Maharashtra	18.32° N	73.51° E	559.0	2002–2023 (22y)

CHAPTER 4: STRATIFIED FRAMEWORK & IMPLEMENTATION

The core breakthrough of this IMD study is the **Seasonal and Regional Stratified Model Selection Framework**. Because solar dynamics are governed by monsoon cycles and coastal proximity, the researchers proved that no single model works everywhere. Instead, a targeted approach must be adopted:

THE SEASON-WISE SELECTION PROTOCOL (IMD RECOMMENDATIONS)

- **1. Inland & Stable Climates (Winter & Post-Monsoon): Use M-1 (Hargreaves–Samani).** Under clear skies, the strong coupling between temperature and radiation makes this simple square-root formulation highly accurate. For instance, in Jabalpur, M-1 achieved an R^2 of 0.66 and a low RMSE of 2.42 MJ/m²/day.
- **2. Monsoon Season Nationwide (June–September): Deploy M-2 (Rao et al.).** Cloud cover decouples diurnal heating from incoming sunlight. The M-2 formulation acts as a noise-resistant model, consistently achieving the highest GPI scores nationwide (e.g., at New Delhi, RMSE was minimized to 4.87 MJ/m²/day).
- **3. Coastal & High-Humidity Environments (All Seasons): Deploy M-5 (Chen) or M-13 (Hassan).** Coastal humidity scatters incoming sunlight, which temperature variations alone fail to reflect. The logarithmic and multi-variable structures of M-5 and M-13 adjust for this moisture-induced scattering (e.g., in Goa and Karaikal, M-5/M-13 minimized RMSE to ~2.0 MJ/m²/day).
- **4. Central Plateaus & Thermally Dynamic Zones: Deploy M-11 or M-13.** During high dust loading or rapid temperature gradients (pre-monsoon), these advanced non-linear formulations outpace simpler models by accounting for dynamic elevation and convective cooling.

PRACTICAL AGRICULTURAL APPLICATIONS & SO-WHAT

- 1. Operational Agrometeorological Advisories:** Standard district-level agromet offices can estimate incoming radiation directly from temperature observations. This enables them to calculate reference crop water requirements (evapotranspiration) and broadcast reliable, daily irrigation advisories to millions of farmers.
- 2. Precision Irrigation Scheduling:** Farmers can optimize water use by matching irrigation volume strictly to computed solar-driven water losses, preventing over-watering and protecting water tables.
- 3. Dynamic Crop Modeling:** Crop growth simulators (like DSSAT or InfoCrop) require daily solar radiation inputs. Using these uncalibrated, region-specific models, researchers can simulate crop yields and evaluate climate change impact in any remote district in India without installing expensive local pyranometers.

STUDY LIMITATIONS & CONCLUSION

Limitations: This study's rankings reflect model performance *without any site-specific calibrations*. Additionally, validation was performed against the Angström-Prescott model proxy rather than direct pyranometric co-location. Future research will focus on high-resolution pyranometric validations and automated machine learning integrations.

Conclusion: This pioneering project shifts the paradigm from applying single generic formulas to a dynamic, seasonally-aware, and geographically-stratified framework, providing an immediately actionable blueprint for sustainable agriculture and climate planning across India.